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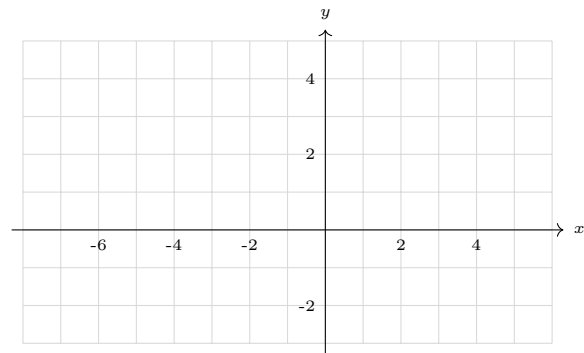
Honors Unit 1 Free Response #1 – Calculator NOT Permitted

Instructions: No calculator. Show all work and justify your reasoning. Each part is scored by the point rubric shown in the answer key.

Consider the piecewise-defined function $f(x)$ and the rational function $g(x)$.

$$f(x) = \begin{cases} -\frac{1}{2}x - 1, & x < -4 \\ 3, & x = -4 \\ -\sqrt{x+4} + 1, & -4 < x \leq 5 \end{cases} \quad g(x) = \frac{x-4}{2x+1}$$

- (4 pts) 1. Determine whether f has a discontinuity at $x = -4$. If it does, classify it. Justify using the behavior of f as $x \rightarrow -4$ from each side.



- (3 pts) 2. Graph $f(x)$. Your graph should confirm your conclusion from the previous part.

- (2 pts) 3. If $f(x) < 0$, what must be true about the graph of f ? State the interval(s) of x on which $f(x) < 0$.

(3 pts) 4. Let $p(x) = -\frac{1}{2}|x + 1|$. For what value(s) of x does $p(x) = f(g(-1))$? Show all work.

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Honors Unit 1 Free Response #2 – Calculator NOT Permitted

Instructions: No calculator. Show all work and justify your reasoning. Each part is scored by the point rubric shown in the answer key.

Consider the piecewise-defined functions $f(x)$ and $g(x)$.

$$f(x) = \begin{cases} x^2 + \frac{2}{3}x, & -9 < x \leq -3 \\ -2x + 1, & -3 < x < 2 \\ x + 3, & x > 2 \end{cases} \quad g(x) = \begin{cases} ax + 3, & x < -2 \\ x^2 + 2x, & x \geq -2 \end{cases}$$

- (4 pts) 1. Find each value or state that it does not exist, and find the domain of f . Show or explain your work.

(1 pts) (a) $f(-9)$

(1 pts) (b) $f(-3)$

(1 pts) (c) $(f + g)(3)$

(1 pts) (d) Domain of f .

- (4 pts) 2. Does f have a discontinuity at $x = -3$? If so, classify it; if not, justify why. Use one-sided behavior.

(3 pts) 3. For what value of a is g continuous at $x = -2$?

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Honors Unit 1 Free Response #3 – Calculator NOT Permitted

Instructions: No calculator. Show all work and justify your reasoning. Each part is scored by the point rubric shown in the answer key.

Let $R(x)$ be a rational function and $S(x)$ a square-root function.

$$R(x) = \frac{x+2}{x+8}, \quad S(x) = \sqrt{x+6} - 4.$$

(2 pts) 1. For what value(s) is $R(x)$ undefined? Explain, and write the domain of R in interval notation.

(2 pts) 2. For what value(s) is $S(x)$ undefined? Explain, and write the domain of S in interval notation.

(3 pts) 3. Let $T(x) = x^2 + x - 14$ and define $h(x) = R(T(x))$.

(1 pts) (a) Write an equation for $h(x)$.

(1 pts) (b) For what values of x is $h(x)$ undefined?

(1 pts) (c) Evaluate $h(3)$ and describe what happens on the graph there.

(3 pts) 4. Let $P(x) = \frac{1}{2}|x - 3|$. Find the value(s) of x (if any) for which $R(-6) \cdot S(3) = P(x)$.

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Honors Unit 1 Free Response #4 – Calculator NOT Permitted

Instructions: No calculator. Show all work and justify your reasoning. Each part is scored by the point rubric shown in the answer key.

The functions $f(x)$ and $g(x)$ are piecewise-defined, consist of connected linear pieces, and each has domain $[-4, 4]$.

x	-4	-1	0	1	4
$f(x)$	3	5	2	5	3
$g(x)$	-3	-2	0	2	3

(2 pts) 1. If $p(x) = x^2 - 3x - 1$, for what value(s) of x is $p(x) = [f(1) - 2 \cdot g(-1)]$? Show your work.

(2 pts) 2. Determine whether f and g are even, odd, or neither. Give a numerical reason for each.

(2 pts) 3. If $p(x) = 2g(-x + 3) - 4$, the point $(-4, -3)$ on the graph of g corresponds to what point on the graph of p ? Explain your reasoning.

(3 pts) 4. If $h(x) = 3f(x) - g(x)$, complete the table of values for $h(x)$. Does $h^{-1}(x)$ exist? Give a reason.

x	-4	-1	0	1	4
$h(x)$	_____	_____	_____	_____	_____

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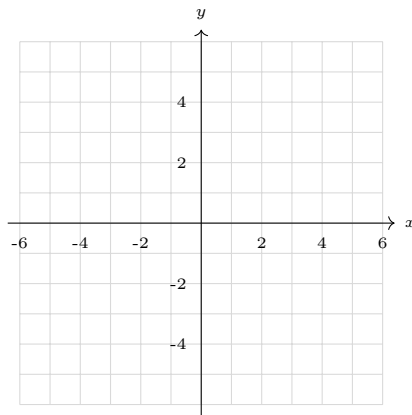
Honors Unit 1 Free Response #5 – Calculator Permitted

Instructions: Show all work and explain your reasoning. Each part is scored by the point rubric shown in the answer key.

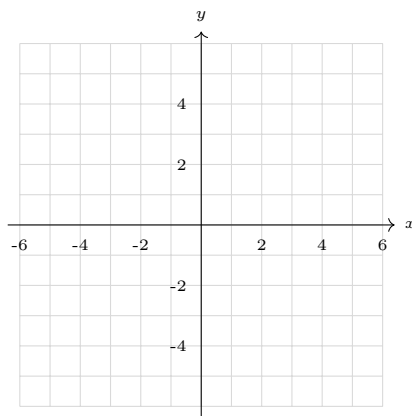
Below are values of a function $f(x)$ on the interval $[0, 6]$. The graph of $f(x)$ consists of line segments connecting the points in the table.

x	0	1	2	4	6
$f(x)$	0	2	2	5	5

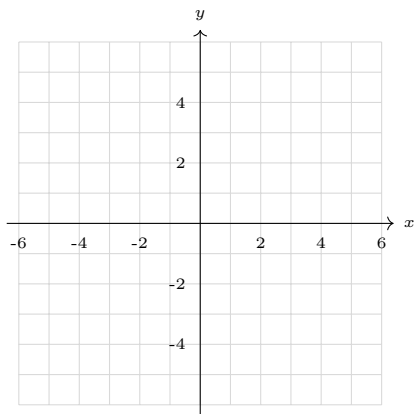
- (3 pts) 1. Suppose $f(-x) = f(x)$. Sketch the graph of $f(x)$ on $[-6, 6]$ and explain your reasoning.



- (3 pts) 2. Sketch $y = -f(x) + 2$ on $[-6, 6]$ (using the even function from the previous part). Explain how you found the new coordinates.



- (3 pts) 3. Suppose instead $f(-x) = -f(x)$. Sketch the graph of $f(x)$ on $[-6, 6]$ and explain your reasoning.



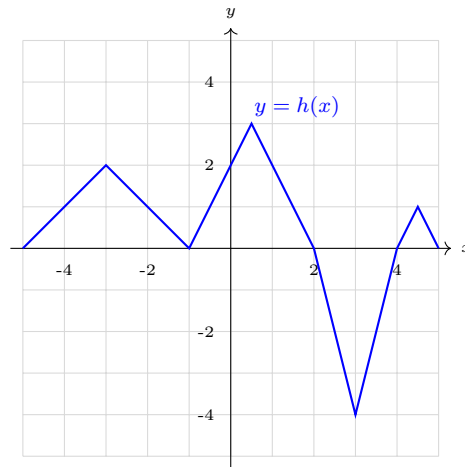
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Honors Unit 1 Free Response #6 – Calculator NOT Permitted

Instructions: No calculator. Show all work and justify your reasoning. Each part is scored by the point rubric shown in the answer key.

The graph of a function $h(x)$ is shown. Another function is defined by $g(x) = -\sqrt{x+5} + 2$.



(2 pts) 1. Identify the interval(s) of x for which $h(x) > 0$. Explain your reasoning.

(3 pts) 2. If $f(x) = |2x + 1| - 4$, for what value(s) of x does $f(x) = g(h(3))$? Show your work.

(2 pts) 3. On the same grid as $h(x)$, graph $g(x)$. For how many values of x does $h(x) = g(x)$? Give a reason.

- (3 pts) 4. Find an equation for $g^{-1}(x)$, if it exists. If it does not, explain why. State any domain restriction.